

CSE110 - Quiz 1 (A) (Time: 25 Minutes)

Name:

ID:

Section:

Draw a flowchart that takes **two integer inputs** from the user: the total purchase amount and the customer type. For the customer type, the system should take **1** for Regular customers and **2** for Premium customers. Based on the purchase amount and customer type, calculate and print the **discount amount** according to the following rules: [Marks: 15]

- If the purchase amount is **greater than or equal to 5000**, apply a **15% discount**.
- Otherwise, apply a **10% discount**.
- If the customer is **Premium (2)**, apply an additional **5% discount**.

Write the outputs of the expressions below in the output boxes. [Your answers won't be accepted without the calculations. [Marks: 5]

1	System.out.println(!false (!true)&&true !false && (5%10==7));
2	System.out.println(6+4*7+3/3.0+7%11-9*2%11);
3	int a = 2, b = 4, c = -9;
4	c = ++b + (--a);
5	System.out.println(c);
6	a = b-- + 5 - c;
7	System.out.println(a);
9	System.out.println(b++);

Output

CSE110 - Quiz 1 (B) (Time: 25 Minutes)

Name:

ID:

Section:

Draw a flowchart that takes **two integer inputs** from the user: the total monthly mobile bill amount and the payment method. For the payment method, the system should take: **1** for Cash payment & **2** for Online payment. Based on the bill amount and payment method, calculate and print the **discount amount** according to the following rules: [Marks: 15]

- If the bill amount is **greater than or equal to 3000**, apply a **12% discount**.
- Otherwise, apply an **8% discount**.
- If the payment method is **Online (2)**, apply an additional **3% discount**.

Write the outputs of the expressions below in the output boxes. [Your answers won't be accepted without the calculations. [Marks: 5]

1	<code>System.out.println(7%3-2*17+10/2+(2%100/4));</code>
2	<code>System.out.println(((true false)&&(false !false))&&!true);</code>
3	<code>int a = -2, b = 12, c = 3;</code>
4	<code>b = --c + (++a);</code>
5	<code>System.out.println(b);</code>
6	<code>c = b -- + 8 - a;</code>
7	<code>System.out.println(a--);</code>
9	<code>System.out.println(c);</code>

Output

CSE110 - Quiz 1 (C) (Time: 25 Minutes)

Name:

ID:

Section:

Draw a flowchart that takes **two integer inputs** from the user: the total parking hours and the vehicle type. For the vehicle type, the system should take: **1** for Car & **2** for Bus. Based on the parking hours and vehicle type, calculate and print the **total parking fee** according to the following rules:

[Marks: 15]

- If parking hours are **greater than or equal to 5**, charge **100 taka per hour**.
- Otherwise, charge **60 taka per hour**.
- If the vehicle type is **Bus (2)**, add an additional **200 taka fixed charge**.

Write the outputs of the expressions below in the output boxes. [Your answers won't be accepted without the calculations. [Marks: 5]

1	System.out.println(!false (!true)&&true !false && (5%10==7));
2	System.out.println(6+4*7+3/3.0+7%11-9*2%11);
3	int a = 2, b = 4, c = -9;
4	c = ++b + (--a);
5	System.out.println(c);
6	a = b-- + 5 - c;
7	System.out.println(a);
9	System.out.println(b++);

Output

CSE110 - Quiz 1 (D) (Time: 25 Minutes)

Name:

ID:

Section:

Draw a flowchart that takes **two integer inputs** from the user: the total overtime hours worked and the employee type. For the employee type, the system should take: **1** for a Junior employee & **2** for a Senior employee. Based on the overtime hours and employee type, calculate and print the **total overtime payment** according to the following rules: [Marks: 15]

- If overtime hours are **greater than or equal to 8**, pay **500 taka per hour**.
- Otherwise, pay **300 taka per hour**.
- If the employee type is **Senior (2)**, add an additional **1000 taka**

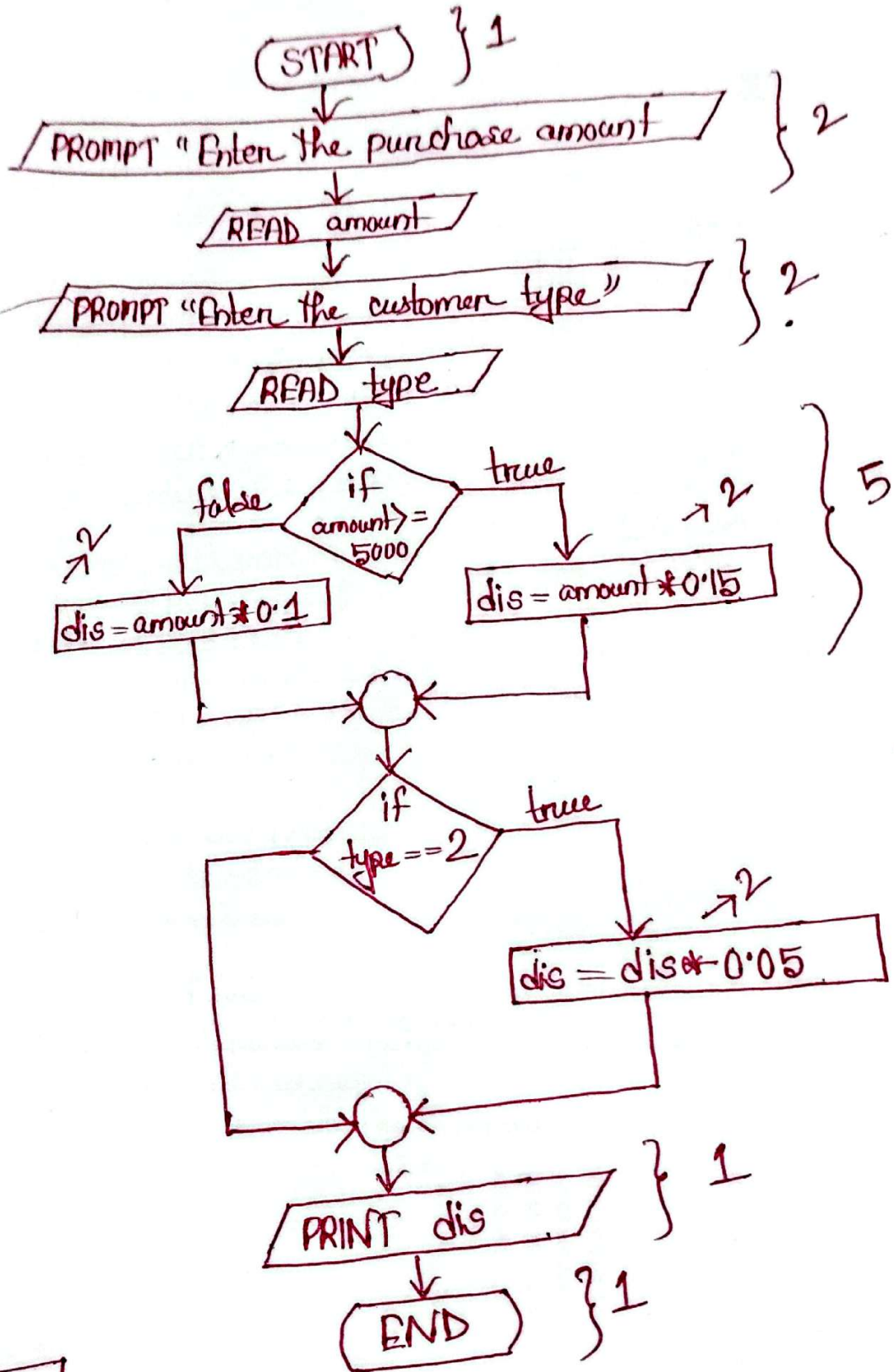
Write the outputs of the expressions below in the output boxes. [Your answers won't be accepted without the calculations. [Marks: 5]

1	System.out.println(7%3-2*17+10/2+(2%100/4));
2	System.out.println(((true false)&&(false !false))&&!true);
3	int a = -2, b = 12, c = 3;
4	b = --c + (++a);
5	System.out.println(b);
6	c = b -- + 8 - a;
7	System.out.println(a--);
9	System.out.println(c);

Output

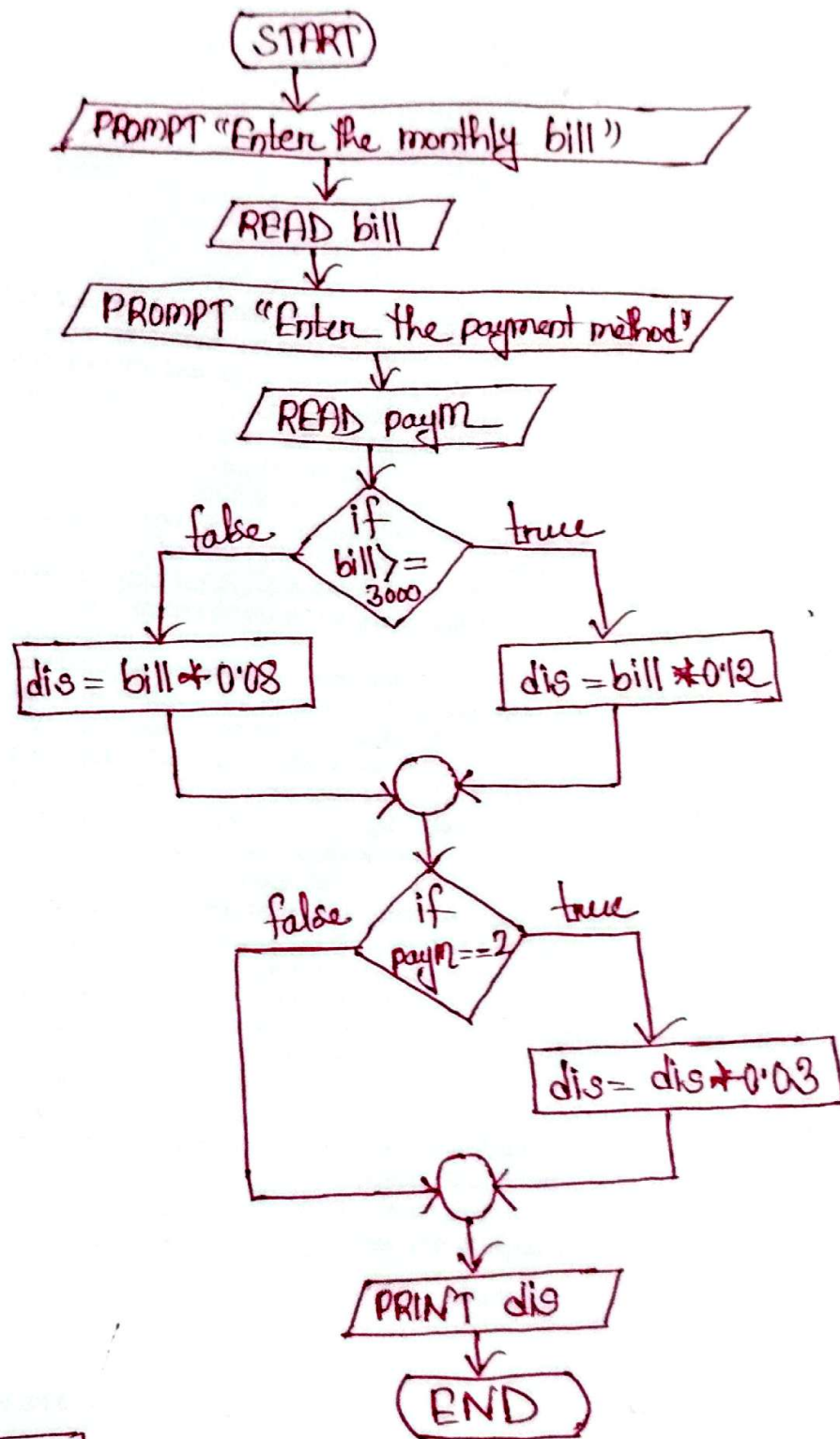
Set-A

1



true
35.0
6
4
4

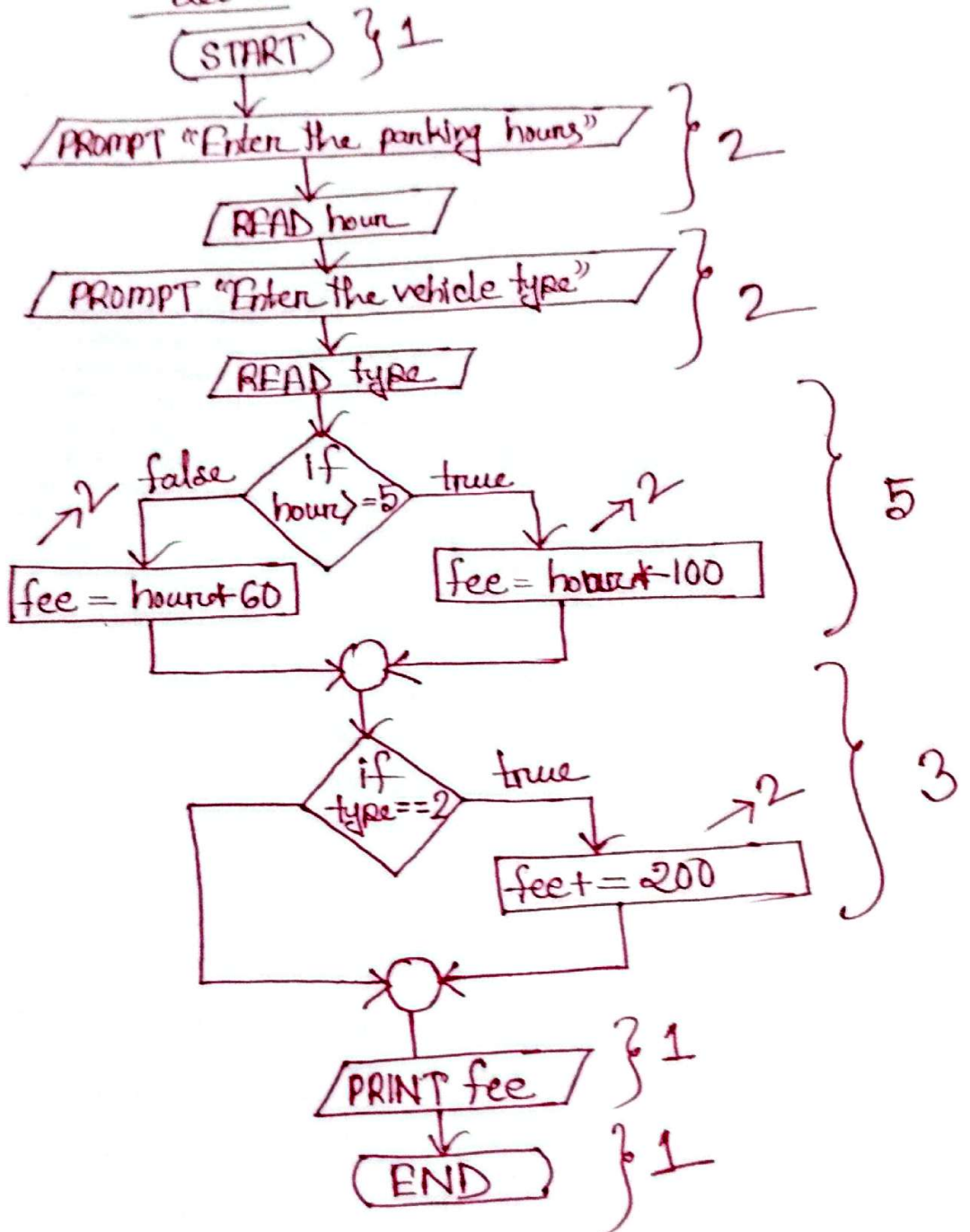
Set B



2

-28
false
1
-1
10

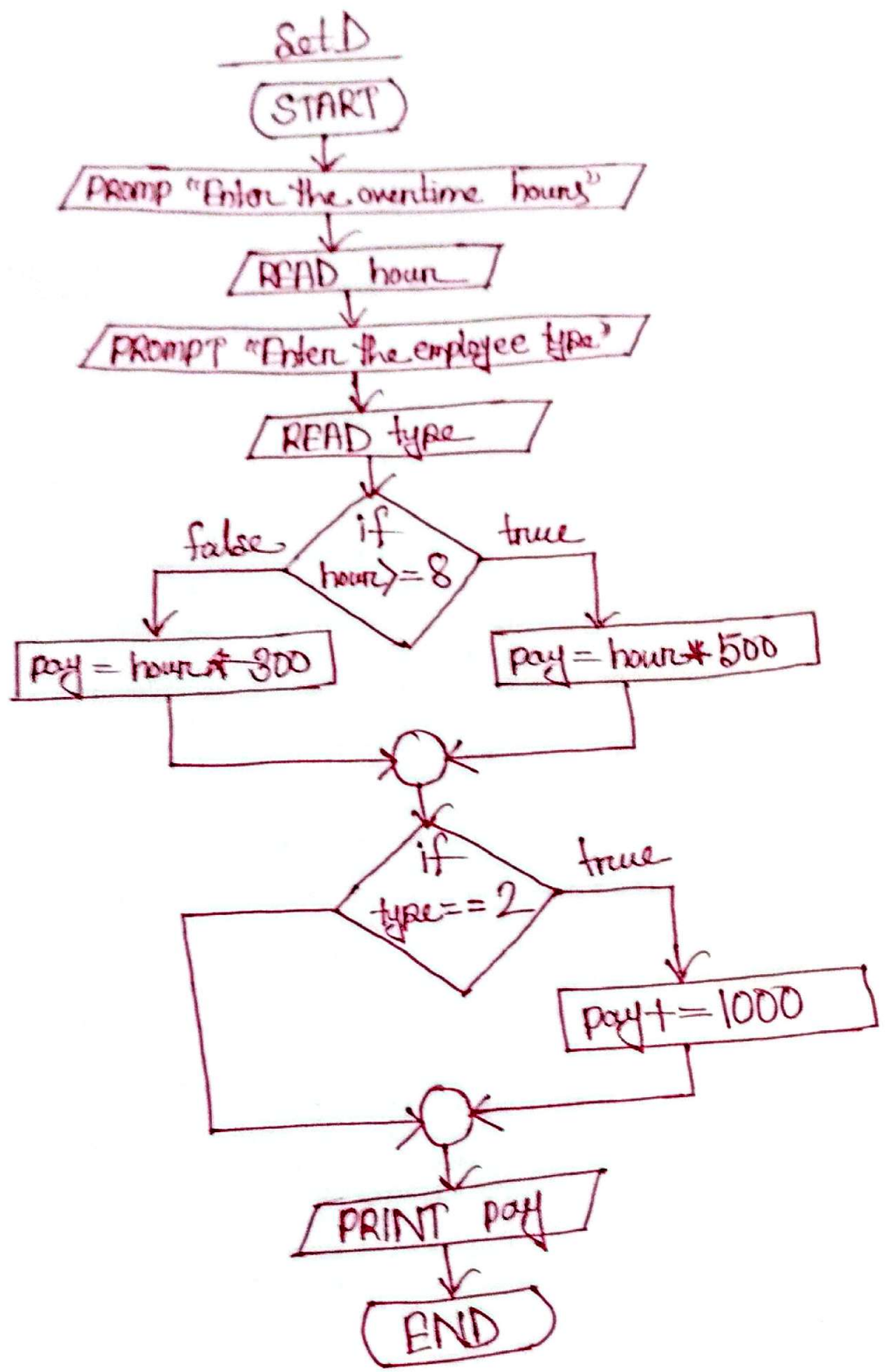
Set C



2

true
35.0
6
4
4

1



2

-28
false
1
-1
10.

A

CSE110 - Quiz 3 (A) (Time: 20 Minutes)**Name:****ID:****Section:**

Write a Java program that will take a String as input from the user and print the summation of the digits that are at odd index. Finally, print the summation. You can't use any other built-in methods other than `.length()` and `.charAt()` methods. [Marks: 10]

Sample Input	Sample Output
AHe6e@61ZUu98	14
34y\$%s2348IY71a	16
Explanation: For sample input 2, The odd indexed digits are 4, 3, 8, and 1. That's why the output is $(4+3+8+1) = 16$	

CSE110 - Quiz 2 (A) (Time: 20 Minutes)

The following code is for printing a Hollow Rectangle. However, it contains 10 errors in 10 different lines. For row = 4 and col = 5, the output shape is also mentioned for your reference. [Marks:10]

1	import java.util.scanner;
2	public class A{
3	public void main(String [] args){
4	Scanner scan = new Scanner(System.out);
5	int row = scan.nextInt();
6	int col = scan.nextInt();
7	for(int r = 1; r <= row-1; r++){
8	for(int c = 0; c <= col; c++){
9	if (r == 1 r == col c == 1 c == col){
10	System.out.print(col);
11	}
12	else{
13	System.out.println(" ");
14	}
15	}
16	System.out.println(r);
17	}
18	}
19	}

12345
1 5
1 5
12345

Please correct the lines and rewrite them in the following boxes.

[illegible]

B

CSE110 - Quiz 2 (B) (Time: 20 Minutes)

Name:

ID:

Section:

Write a Java program that will take a String as input from the user and print the summation of the digits that are at even index. Finally, print the summation. You can't use any other built-in methods other than `.length()` and `.charAt()` methods. [Marks: 10]

Sample Input	Sample Output
AHe6e@51ZUu98	13
94y\$%s2348IY71a	22
Explanation: For sample input 2, The even-indexed digits are 9, 2, 4, and 7. That's why the output is $(9+2+4+7) = 22$	

CSE110 - Quiz 2 (B) (Time: 20 Minutes)

The following code is for printing a Hollow Triangle. However, it contains 10 errors in 10 different lines. For $n = 5$, the output shape is also mentioned for your reference. [Marks:10]

1	import java.Util.Scanner;
2	public class B{
3	public static main(String[] args) {
4	Scanner sc = Scanner(System.in);
5	double n = sc.nextInt();
6	for (int i = 1; i <= n-1; i++) {
7	for (int j = 0; j <= i; j++) {
8	if (i == 1 i != n j == 1 j == i){
9	System.out.print(i);
10	}
11	else{
12	System.out.println(" ");
13	}
14	}
15	System.out.println(n);
16	}
17	}
18	}

1
12
1 3
1 4
12345

Please correct the lines and rewrite them in the following boxes.

[illegible]

C

CSE110 - Quiz 2 (C) (Time: 20 Minutes)**Name:****ID:****Section:**

Write a Java program that will take a String as input from the user and print the summation of the digits that are at index divisible by 3. Finally, print the summation. You can't use any other built-in methods other than `.length()` and `.charAt()`. [Marks: 10]

Sample Input	Sample Output
AHe7e@61ZUu98	21
34y\$%s2348IYz1a	13
Explanation: For sample input 2, the digits of index divisible by 3 (0, 6, 9) are 3, 2, and 8. That's why the output is $(3+2+8) = 13$	

CSE110 - Quiz 2 (C) (Time: 20 Minutes)

The following code is for printing a Hollow Rectangle. However, it contains 10 errors in 10 different lines. For row = 4 and col = 5, the output shape is also mentioned for your reference. [Marks:10]

1	import java.util.scanner;
2	public class A{
3	public void main(String [] args){
4	Scanner scan = new Scanner(System.out);
5	int row = scan.nextInt();
6	int col = scan.nextInt();
7	for(int r = 1; r <= row-1; r++){
8	for(int c = 0; c <= col; c++){
9	if (r == 1 r == col c == 1 c == col){
10	System.out.print(col);
11	}
12	else{
13	System.out.println(" ");
14	}
15	}
16	System.out.println(r);
17	}
18	}
19	}

12345
1 5
1 5
12345

Please correct the lines and rewrite them in the following boxes.

[illegible]

D

CSE110 - Quiz 2 (D) (Time: 20 Minutes)

Name:

ID:

Section:

Write a Java program that will take a String as input from the user and print the summation of the digits that are at index divisible by 4. Finally, print the summation. You can't use any other built-in methods other than `.length()` and `.charAt()`. [Marks: 10]

Sample Input	Sample Output
2He7e@61ZUu98	10
34y\$5s2348IYz1a	12
Explanation: For sample input 2, the digits of index divisible by 4 (0, 4, 8) are 3, 5, and 4. That's why the output is $(3+5+4) = 12$	

CSE110 - Quiz 2 (D) (Time: 20 Minutes)

The following code is for printing a Hollow Triangle. However, it contains 10 errors in 10 different lines. For $n = 5$, the output shape is also mentioned for your reference. [Marks:10]

1	import java.Util.Scanner;
2	public class B{
3	public static main(String[] args) {
4	Scanner sc = Scanner(System.in);
5	double n = sc.nextInt();
6	for (int i = 1; i <= n-1; i++) {
7	for (int j = 0; j <= i; j++) {
8	if (i == 1 i != n j == 1 j == i){
9	System.out.print(i);
10	}
11	else{
12	System.out.println(" ");
13	}
14	}
15	System.out.println(n);
16	}
17	}
18	}

1
12
1 3
1 4
12345

Please correct the lines and rewrite them in the following boxes.

[illegible]

Solution

SET A:

```
import java.util.Scanner;
public class A{
    public static void main(String [] args){
        Scanner sc = new Scanner(System.in);
        String s = sc.nextLine();
        int num = 0;
        for(int i = 0; i<s.length(); i++){
            if(i%2==0){
                char ch = s.charAt(i);
                if(ch>='0' && ch<='9'){
                    num+= (int) (ch-'0');
                }
            }
        }
        System.out.println(num);
    }
}
```

SET B:

```
import java.util.Scanner;
public class A{
    public static void main(String [] args){
        Scanner sc = new Scanner(System.in);
        String s = sc.nextLine();
        int num = 0;
        for(int i = 0; i<s.length(); i++){
            if(i%2!=0){
                char ch = s.charAt(i);
                if(ch>='0' && ch<='9'){
                    num+= (int) (ch-'0');
                }
            }
        }
        System.out.println(num);
    }
}
```

SET C:

```
import java.util.Scanner;
public class A{
    public static void main(String [] args){
        Scanner sc = new Scanner(System.in);
        String s = sc.nextLine();
        int num = 0;
        for(int i = 0; i<s.length(); i++){
            if(i%3==0){
                char ch = s.charAt(i);
                if(ch>='0' && ch<='9'){
                    num+= (int) (ch-'0');
                }
            }
        }
        System.out.println(num);
    }
}
```

SET D:

```
import java.util.Scanner;
public class A{
    public static void main(String [] args){
        Scanner sc = new Scanner(System.in);
        String s = sc.nextLine();
        int num = 0;
        for(int i = 0; i<s.length(); i++){
            if(i%4==0){
                char ch = s.charAt(i);
                if(ch>='0' && ch<='9'){
                    num+= (int) (ch-'0');
                }
            }
        }
        System.out.println(num);
    }
}
```

SET: A

1	import java.util. scanner ;
3	public static void main(String [] args){
4	Scanner scan = new Scanner(System.in);
5	int row = scan. nextInt ();
7	for(int r = 1; r <= row ; r++){
8	for(int c = 1 ; c <= col; c++){
9	if (r == 1 r == row c == 1 c == col){
10	System.out.print(c);
13	System.out. print (" ");
16	System.out. println ();

SET: B

1	import java. Util .Scanner;
3	public static void main(String [] args){
4	Scanner scan = new Scanner(System.out);
5	int n = sc.nextInt();
6	for (int i = 1; i <= n ; i++) {
7	for (int j = 1 ; j <= i; j++) {
8	if (i == 1 i == n j == 1 j == i){
9	System.out. print (j);
12	System.out. print (" ");
15	System.out. println ();

SET C:

1	import java.util. scanner ;
3	public static void main(String [] args){

4	Scanner scan = new Scanner(System.in);
5	int row = scan.nextInt();
7	for(int r = 1; r <= row ; r++){
8	for(int c = 1 ; c <= col; c++){
9	if (r == 1 r == row c == 1 c == col){
10	System.out.print(c);
13	System.out. print (" ");
16	System.out. println ();

SET D:

1	import java. Util .Scanner;
3	public static void main(String [] args){
4	Scanner scan = new Scanner(System.out);
5	int n = sc.nextInt();
6	for (int i = 1; i <= n ; i++) {
7	for (int j = 1 ; j <= i; j++) {
8	if (i == 1 i == n j == 1 j == i){
9	System.out. print (j);
12	System.out. print (" ");
15	System.out. println ();